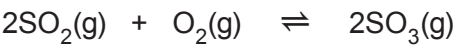


Examiner
only

5. (a) Sulfuric acid is manufactured using the Contact Process. The equation represents one stage of the process.



Describe and explain **all** the stages in the process, including the one given above, from **raw materials** to the final product, **sulfuric acid**. [6 QER]

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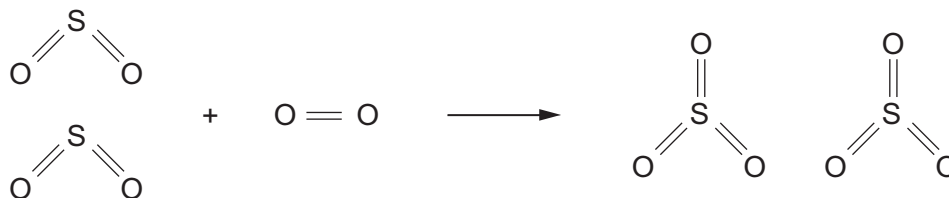
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- (b) The equation shows the bonds which are broken and the bonds which are formed in the production of sulfur trioxide.



The bond energy of an $\text{S}=\text{O}$ bond is 523 kJ.

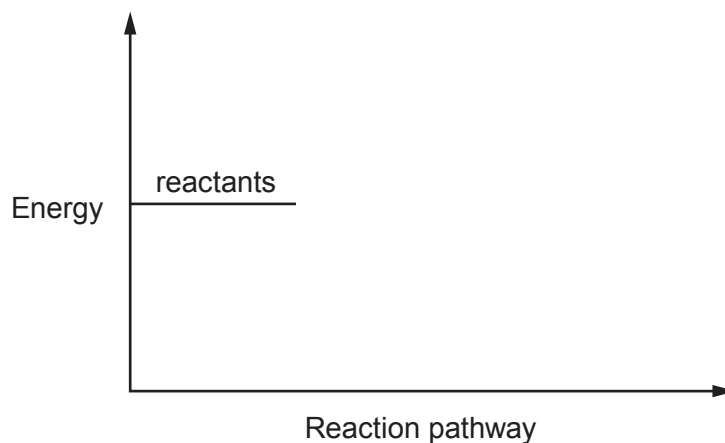
- (i) The **total** energy needed to break the bonds in the reactants is 2587 kJ. Calculate the energy needed to break an $\text{O}=\text{O}$ bond. [2]

Energy = kJ

- (ii) Calculate the overall energy change for the reaction. [2]

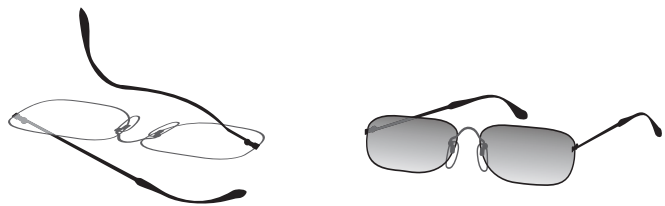
Overall energy change = kJ

- (iii) Complete the energy profile for the reaction. [1]



Examiner
only

5. (a) Smart materials are used to make the frames and lenses of certain spectacles.



Give the names of the different types of smart material used. Describe the unusual property of each. [2]

Frames

.....

.....

Lenses

.....

.....



Examiner
only

(b) Titanium dioxide has been used in sun creams for decades. Sun screens available today use nano-scale titanium dioxide particles. Some people believe using creams containing nano-particles is unsafe.

(i) Give the advantage of using nano-scale titanium dioxide particles rather than larger titanium dioxide particles to make sun screens. [1]

.....

.....

(ii) Explain why some people are concerned about the use of nano-scale titanium dioxide particles in sun screens. [2]

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(iii) **Approximately** how many times bigger are common titanium dioxide particles ($3 \times 10^{-7}\text{m}$) than nano-scale titanium dioxide particles ($2.5 \times 10^{-10}\text{m}$)? [2]

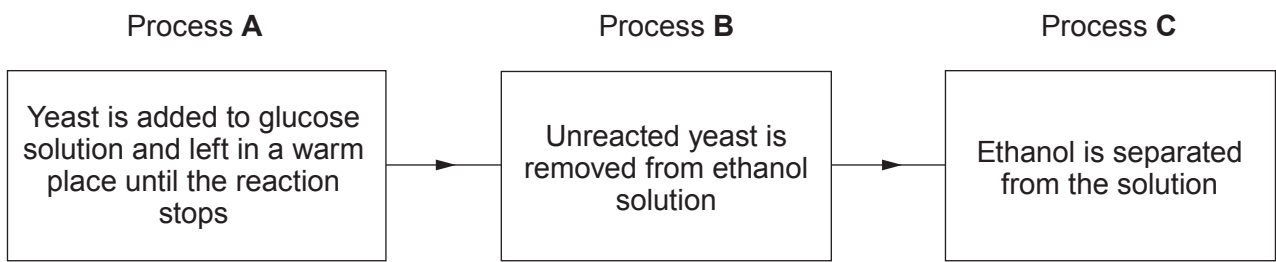
Answer =

7

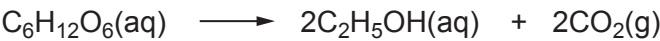
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5. The chart describes the laboratory preparation of ethanol from glucose solution.



The equation for the reaction occurring in process A is shown below.



(a) (i) Give the name of the processes A, B and C. [3]

- A
- B
- C



(ii) A teacher wanted to show that ethanol is collected in process **C**.

- I. Tick (✓) the box next to the chemical test that the teacher would carry out to positively identify the liquid as ethanol. [1]

add bromine water

☐

add acidified potassium dichromate solution

☐

add silver nitrate solution

☐

add barium chloride solution

☐

- II. Tick (✓) the box next to the observation you would expect. [1]

orange to colourless

☐

orange to green

☐

green to orange

☐

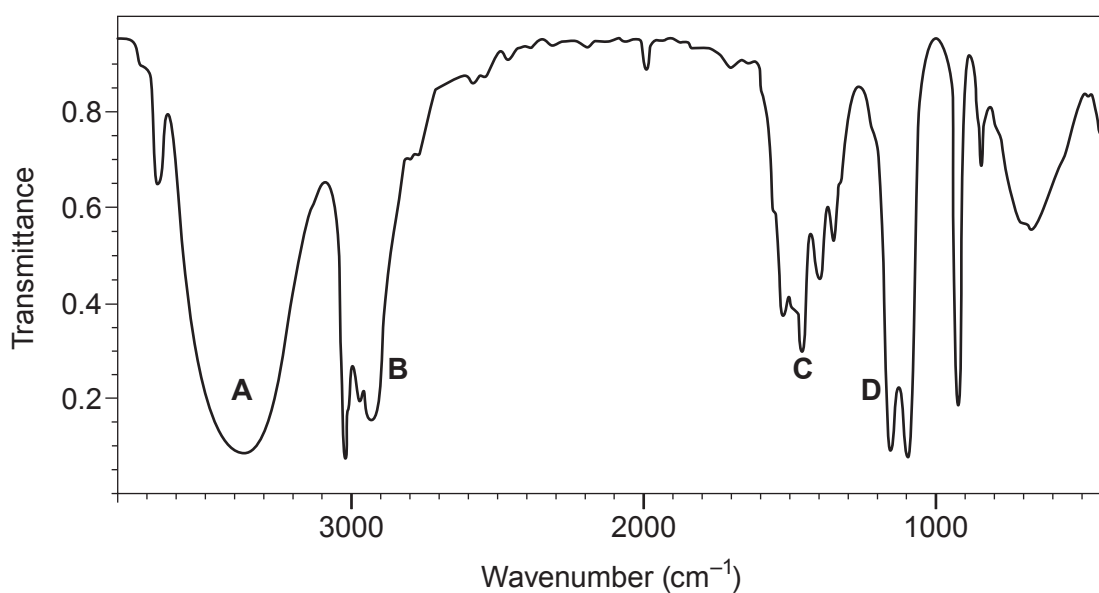
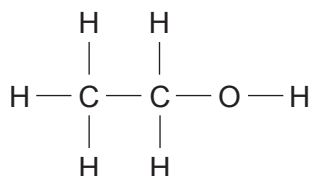
colourless to green

☐

(b) Alcohols can also be identified using infrared spectroscopy.

Bond	Wavenumber (cm^{-1})
$\text{C}=\text{C}$	1620 to 1670
$\text{C}=\text{O}$	1650 to 1750
$\text{C}-\text{H}$	2800 to 3100
$\text{O}-\text{H}$	2500 to 3550

(i) The structural formula and the infrared spectrum of ethanol are shown.



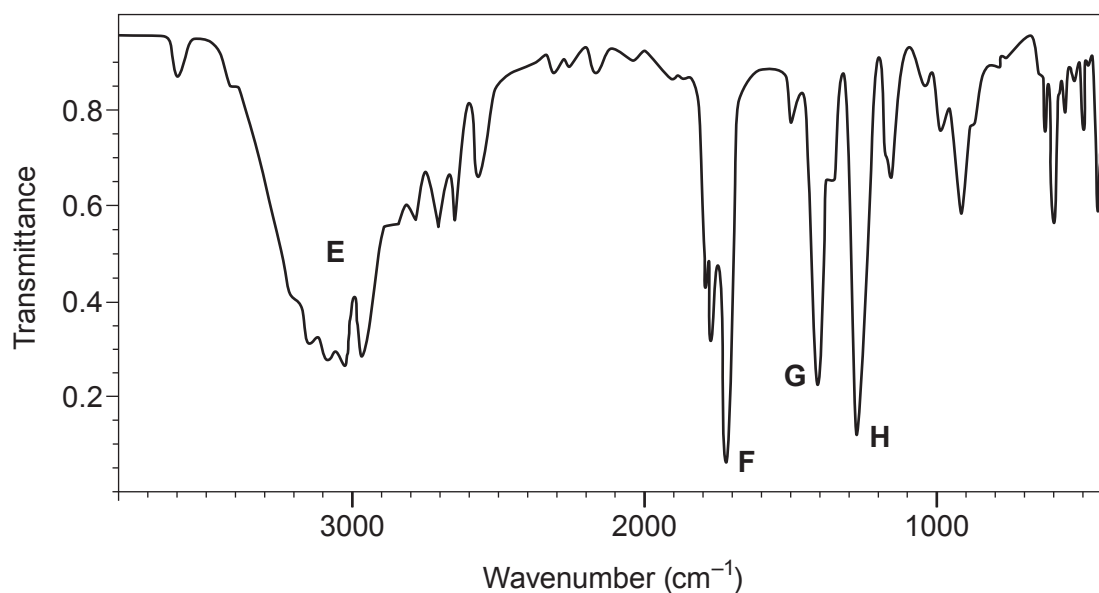
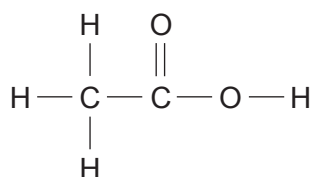
Give the **letter** of the peak that can be used to identify an alcohol.

[1]

Letter



- (ii) The structural formula and the infrared spectrum of ethanoic acid are shown.



Give the **letter** of the peak that can be used to distinguish ethanoic acid from ethanol.

[1]

Letter

- (c) Ethanol is found in alcoholic drinks.

- (i) Give **one** health problem associated with alcohol abuse over a **long** period of time.

[1]

.....

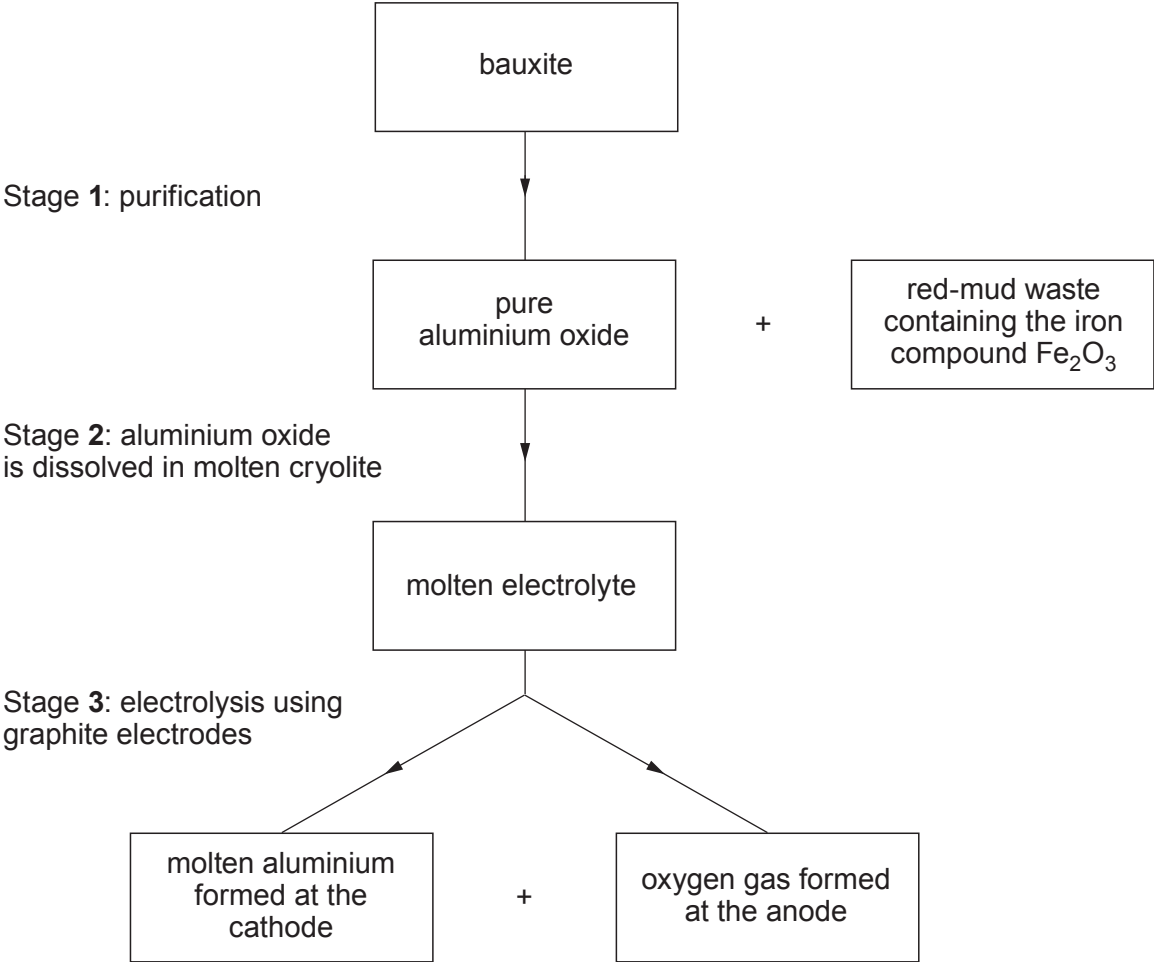
- (ii) Give **one** social problem associated with the excessive intake of alcohol (binge drinking) during an evening.

[1]

.....



5. (a) The flow chart shows the stages involved in the extraction of aluminium from bauxite. Bauxite is a mixture of mainly aluminium oxide and a compound of iron.



- (i) Underline the chemical name for the iron compound, Fe_2O_3 , found in the red-mud waste. [1]

iron(V) oxide

iron(II) oxide

iron(III) oxide

iron(I) oxide

- (ii) Give **one** reason why aluminium oxide is dissolved in molten cryolite in stage 2. [1]

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.....

- (iii) Explain why the anode is regularly replaced in stage 3. [2]

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.....

- (iv) Complete and balance the equation for the electrolysis of aluminium oxide. [2]



- (b) The recycling rate for aluminium drinks cans in the UK reached 72 % in 2017.

Suggest why most aluminium waste is recycled. [2]

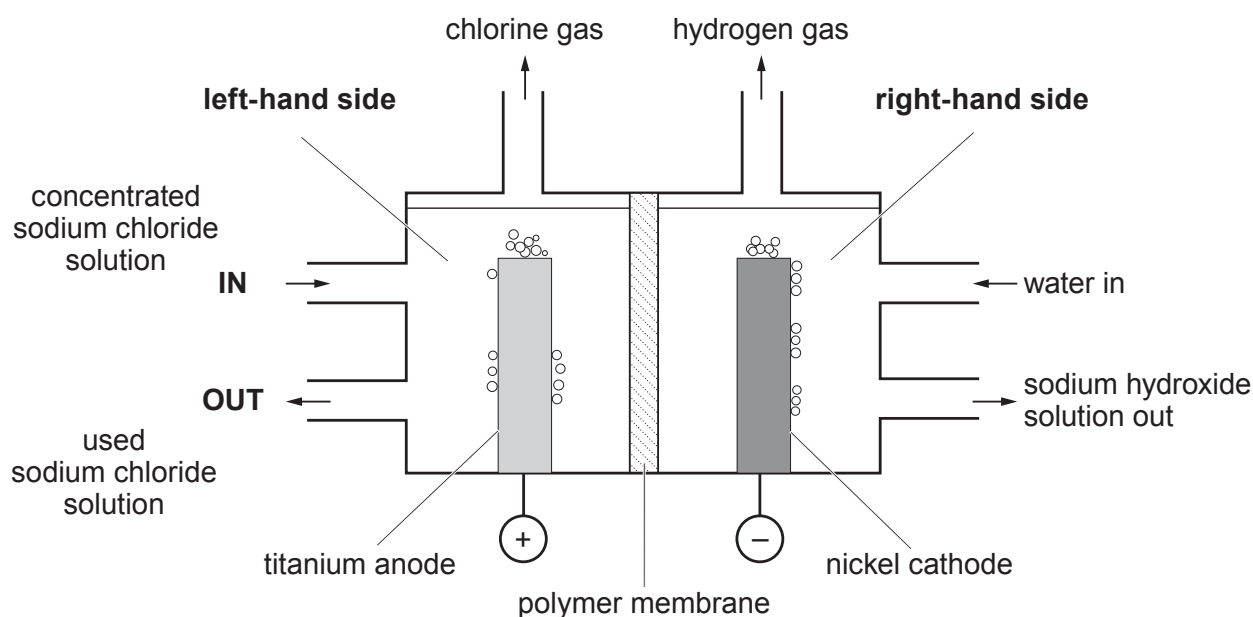
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- (c) The diagram shows a model of a membrane cell used to manufacture sodium hydroxide solution, chlorine and hydrogen by the electrolysis of aqueous sodium chloride.

The membrane cell is designed so that chlorine is kept separate from hydrogen and aqueous sodium hydroxide.

The polymer membrane only allows positive ions to pass through it. Sodium ions and hydrogen ions pass through from the left-hand side to the right-hand side of the cell.



Chlorine is an extremely corrosive substance which reacts with most metals. Titanium reacts quickly with chlorine gas but it resists attack by chlorine in aqueous conditions.



- (i) Put a **tick (✓)** in the box next to the correct list of ions found on either side of the membrane during the process. [1]

Left-hand side of the membrane	Right-hand side of the membrane	
Na ⁺ Cl ⁻ H ⁺ OH ⁻	Cl ⁻ H ⁺ OH ⁻	☐
H ⁺ OH ⁻ Cl ⁻	Na ⁺ H ⁺ OH ⁻	☐
Na ⁺ Cl ⁻ H ⁺ OH ⁻	Na ⁺ H ⁺ OH ⁻	☐
Na ⁺ Cl ⁻	Na ⁺ OH ⁻	☐

- (ii) Put a **tick (✓)** in the box next to the correct statement. [1]

- titanium burns in chlorine in aqueous conditions ☐
- titanium doesn't react with chlorine under any conditions ☐
- aqueous conditions prevent chlorine from reacting with titanium ☐
- aqueous conditions prevent chlorine from reacting with sodium chloride ☐

- (iii) Put a **tick (✓)** in the box next to the two substances formed on the right-hand side of the membrane if the sodium chloride solution were contaminated with magnesium bromide. [1]

- sodium hydroxide and magnesium hydroxide ☐
- sodium hydroxide and sodium bromide ☐
- sodium hydroxide and magnesium bromide ☐
- magnesium chloride and sodium ☐



Examiner
only

5. (a) (i) Draw a dot and cross diagram to show the bonding in a carbon dioxide molecule. [2]

carbon (2,4) oxygen (2,6)

(ii) Explain why carbon dioxide has a low boiling point. [2]

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Examiner
only

- (b) (i) Draw a diagram to show the electronic changes that take place during the formation of magnesium oxide. Include the charges on the ions formed. [2]

magnesium (2,8,2) oxygen (2,6)

- (ii) Explain why magnesium oxide has a higher melting point than sodium chloride. [2]

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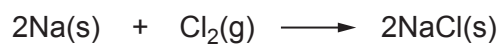
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- (iii) Sodium chloride is produced by the reaction of sodium with chlorine.



In a reaction using 0.080 mol of sodium, 4.12 g of sodium chloride was produced.

Calculate the percentage yield of this reaction. [3]

$$A_r(\text{Na}) = 23$$

$$A_r(\text{Cl}) = 35.5$$

Percentage yield = %

